

Appendix A

Standard ODB++ Symbols

A symbol is one of the geometric primitives supported by ODB++ described here, or a combination of them. They are defined by the parameters: width, height, radius, diameter, spokes, gap, angle, size and corner. The corners of rounded or chamfered rectangles or thermals are specified in ascending order—moving counter-clockwise—starting from the top-right corner.

For an example of a symbol with chamfered corners, see [“Rounded/Chamfered Rectangles”](#) on page 203.

All symbols described in this section are standard (system) symbols. User defined symbols are defined as described in [“symbols \(User-Defined Symbols\)”](#) on page 92.

[Basic Standard Symbols](#)

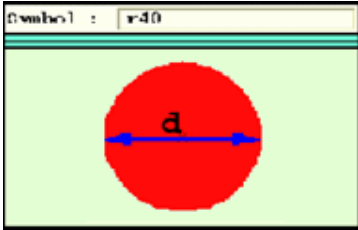
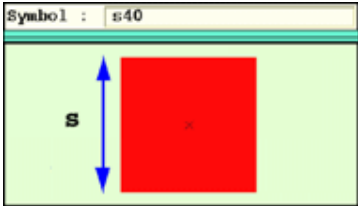
[Symbols Suitable for Solder Stencil Design](#)

[Other Symbol Information](#)

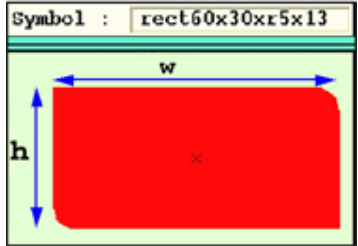
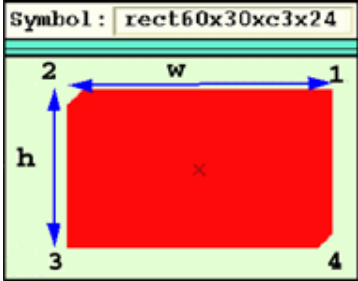
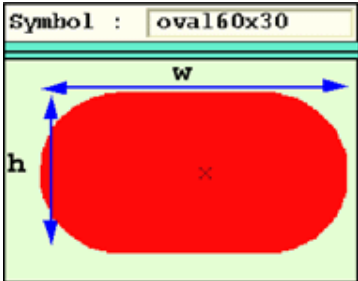
[Obsolete Symbols](#)

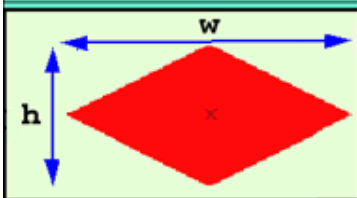
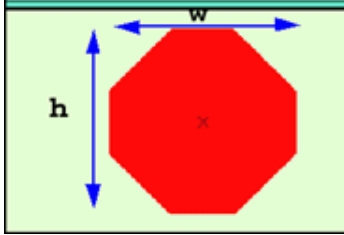
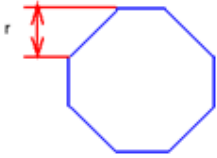
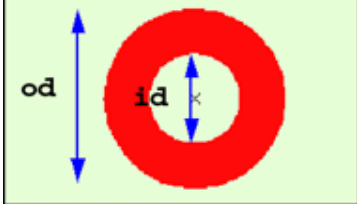
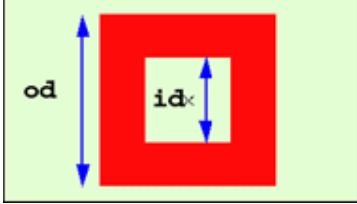
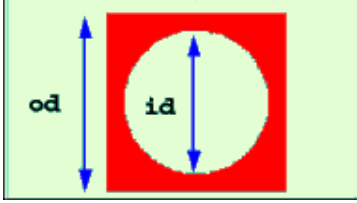
Basic Standard Symbols

These are the basic standard symbols.

Symbol Name	Example	Parameters
Round		r<d> d — Circle diameter
Square		s<s> s — Square side

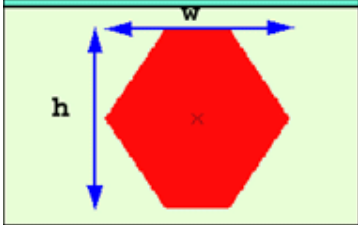
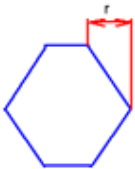
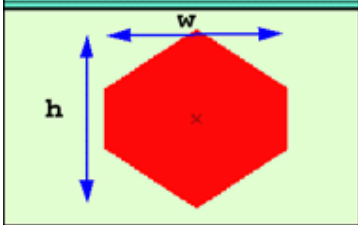
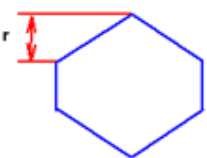
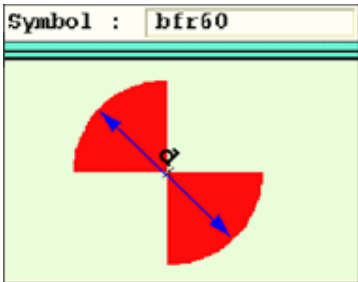
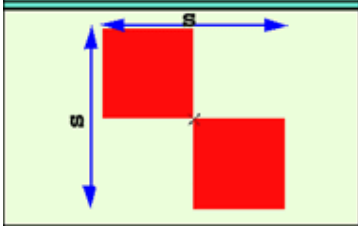
Standard ODB++ Symbols
Basic Standard Symbols

Symbol Name	Example	Parameters
Rectangle		rect<w>x<h> w — Rectangle width h — Rectangle height
Rounded Rectangle		rect<w>x<h>xr<rad>x<corners> w — Rectangle width h — Rectangle height rad — Corner radius corners — Indicates which corners are rounded. x<corners> is omitted if all corners are rounded.  See “Rounded/Chamfered Rectangles” on page 203.
Chamfered Rectangle		rect<w>x<h>xc<rad>x<corners> w — Rectangle width h — Rectangle height rad — Corner radius corners — Indicates which corners are chamfered. x<corners> is omitted if all corners are chamfered.  See “Rounded/Chamfered Rectangles” on page 203.
Oval		oval<w>x<h> w — Oval width h — Oval height

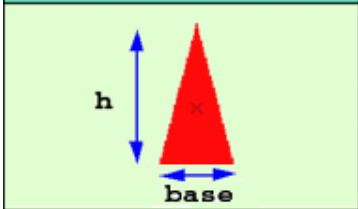
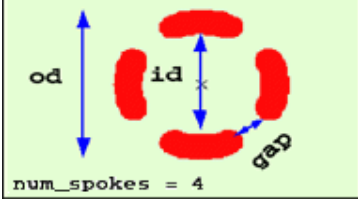
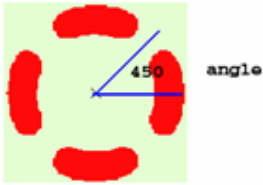
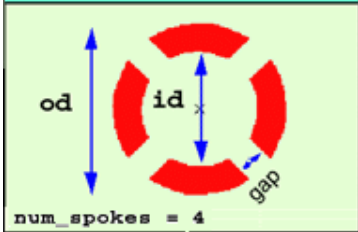
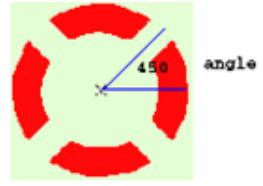
Symbol Name	Example	Parameters
Diamond	Symbol : <code>di60x30</code> 	di<w>x<h> w — Diamond width h — Diamond height
Octagon	Symbol : <code>oct60x60x20</code> 	oct<w>x<h>x<r> w — Octagon width h — Octagon height r — Corner size 
Round Donut	Symbol : <code>donut_r60x30</code> 	donut_r<od>x<id> od — Outer diameter id — Inner diameter
Square Donut	Symbol : <code>donut_s60x30</code> 	donut_s<od>x<id> od — Outer diameter id — Inner diameter
Square / Round Donut	Symbol : <code>donut_sr10x8</code> 	donut_sr<od>x<id> od — Outer diameter id — Inner diameter

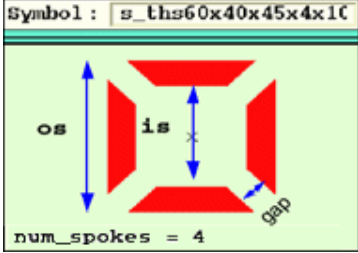
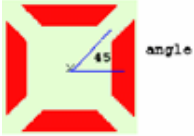
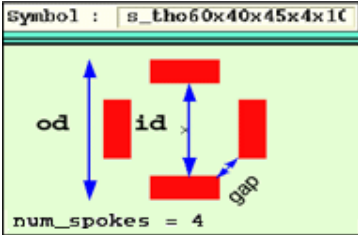
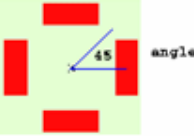
Standard ODB++ Symbols
Basic Standard Symbols

Symbol Name	Example	Parameters
Rounded Square Donut		donut_s<od>x<id>xr<rad>x<corners> od — Outer diameter id — Inner diameter rad — Corner radius corners — Indicates which corners are rounded. x<corners> is omitted if all corners are rounded.
Rectangle Donut		donut_rc<ow>x<oh>x<lw> ow — Outer width oh — Outer height lw — Line width
Rounded Rectangle Donut		donut_rc<ow>x<oh>x<lw>xr<rad>x<corners> ow — Outer width oh — Outer height lw — Line width rad — Corner radius corners — Indicates which corners are rounded. x<corners> is omitted if all corners are rounded.
Oval Donut		donut_o<ow>x<oh>x<lw> ow — Outer width oh — Outer height lw — Line width

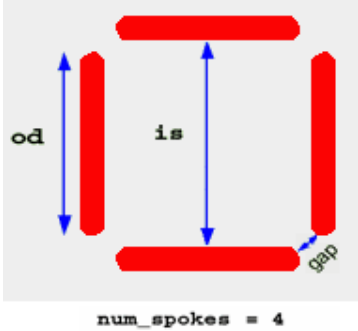
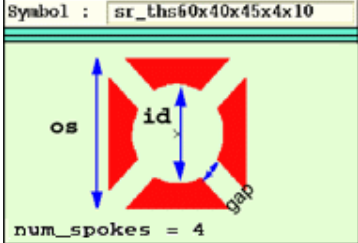
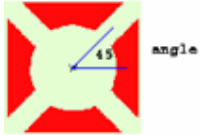
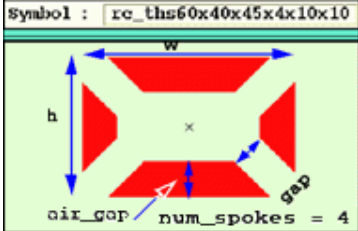
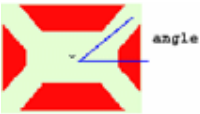
Symbol Name	Example	Parameters
Horizontal Hexagon	Symbol : <code>hex_160x60x20</code> 	<code>hex_l<w>x<h>x<r></code> w — Hexagon width h — Hexagon height r — Corner size 
Vertical Hexagon	Symbol : <code>hex_s60x60x20</code> 	<code>hex_s<w>x<h>x<r></code> w — Hexagon width h — Hexagon height r — Corner size 
Butterfly	Symbol : <code>bfr60</code> 	<code>bfr<d></code> d — Diameter
Square Butterfly	Symbol : <code>bfs60</code> 	<code>bfs<s></code> s — Size

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Symbol Name	Example	Parameters
Triangle	Symbol : tri30x60 	tri<base>x<h> base — Triangle base h — Triangle height
Half Oval	Symbol : oval_h60x30 	oval_h<w>x<h> w — Width h — Height
Round Thermal (Rounded)	Symbol : thr60x40x45x4x10 	thr<od>x<id>x<angle>x<num_spokes>x<gap> od — Outer diameter id — Inner diameter angle — Gap angle from 0 degrees  num_spokes — Number of spokes gap — Size of spoke gap od and id control the air gap (size of laminate separation).
Round Thermal (Squared)	Symbol : ths60x40x45x4x10 	ths<od>x<id>x<angle>x<num_spokes>x<gap> od — Outer diameter id — Inner diameter angle — Gap angle from 0 degrees 

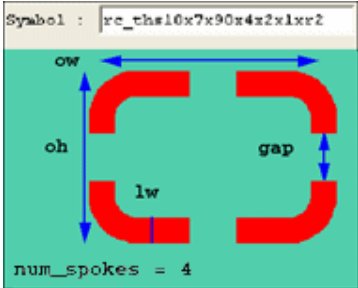
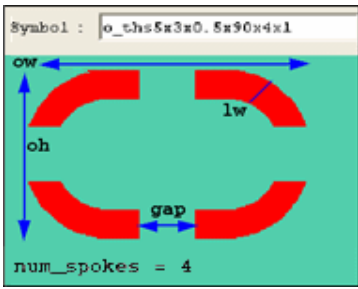
Symbol Name	Example	Parameters
		num_spokes — Number of spokes gap — Size of spoke gap od and id control the air gap (size of laminate separation).
Square Thermal		s_ths<os>x<is>x<angle>x<num_spokes>x<gap> os — Outer size is — Inner size angle — Gap angle from 0 degrees  num_spokes — Number of spokes gap — Size of spoke gap os and is control the air gap (size of laminate separation).
Square Thermal (Open Corners)		s_tho<od>x<id>x<angle>x<num_spokes>x<gap> od — Outer diameter id — Inner diameter angle — Gap angle from 0 degrees in increments of 45 degrees  num_spokes — Number of spokes: 1, 2, or 4 gap — Size of spoke gap od and id control the air gap (size of laminate separation).

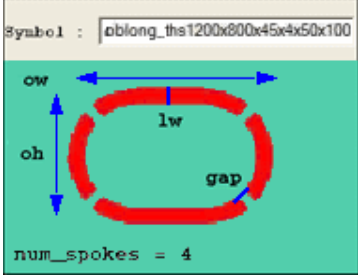
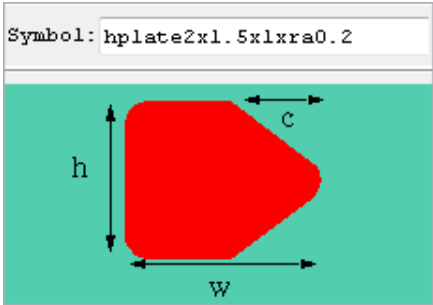
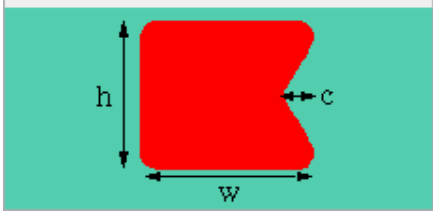
Standard ODB++ Symbols
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Symbol Name	Example	Parameters
Line Thermal		s_thr<os>x<is>x<angle>x<num_spokes>x<gap> os — Outer size is — Inner size angle — Gap angle always 45 degrees num_spoke — number of spokes always 4 gap — Size of spoke gap Ends of lines are rounded with diameter (os-is)/2.
Square-Round Thermal	Symbol : sr_ths60x40x45x4x10 	sr_ths<os>x<id>x<angle>x<num_spokes>x<gap> os — Outer size id — Inner diameter angle — Gap angle from 0 degrees  num_spokes — Number of spokes gap — Size of spoke gap os and id control the air gap (size of laminate separation).
Rectangular Thermal	Symbol : rc_ths60x40x45x4x10x10 	rc_ths<w>x<h>x<angle>x<num_spokes>x<gap>x<air_gap> w — Outer width h — Outer height angle — Gap angle from 0 degrees; in multiples of 45 degrees  num_spokes — Number of spokes gap — Size of spoke gap air_gap — Size of laminate separation

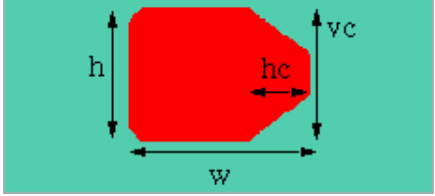
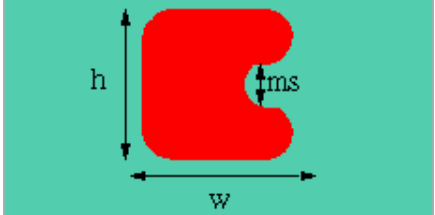
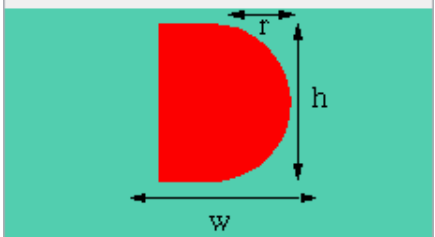
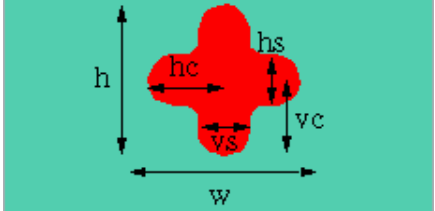
Symbol Name	Example	Parameters
Rectangular Thermal (Open Corners)		rc_tho<w>x<h>x<angle>x<num_spokes>x<gap>x<air_gap> w — Outer width h — Outer height angle — Gap angle from 0 degrees in multiples of 45 degrees. num_spokes — Number of spokes gap — Size of spoke gap air gap — Size of laminate separation
Rounded Square Thermal		s_ths<os>x<is>x<angle>x<num_spokes>x<gap>x<rad>x<corners> os — Outer size is — Inner size angle — Gap angle from 0 degrees num_spokes — Number of spokes gap — Size of spoke gap rad — Corner radius corners — Indicates which corners are rounded. x<corners> is omitted if all corners are rounded.
Rounded Square Thermal (Open Corners)		s_ths<os>x<is>x<angle>x<num_spokes>x<gap>x<rad>x<corners> os — Outer size is — Inner size angle — Gap angle from 45 degrees

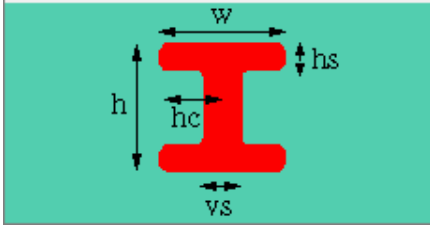
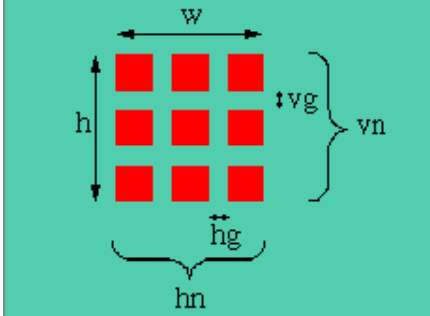
Standard ODB++ Symbols
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Symbol Name	Example	Parameters
		num_spokes — Number of spokes gap — Size of spoke gap rad — Corner radius corners — Indicates which corners are rounded. x<corners> is omitted if all corners are rounded. 
Rounded Rectangle Thermal		rc_ths<ow>x<oh>x<angle>x<num_spokes>x<gap>x<lw>x<rad>x<corners> ow — Outer width oh — Outer height lw — Line width angle — Gap angle from 0 degrees  num_spokes — Number of spokes gap — Size of spoke gap rad — Corner radius corners — Indicates which corners are rounded. x<corners> is omitted if all corners are rounded. 
Oval Thermal		o_ths<ow>x<oh>x<angle>x<num_spokes>x<gap>x<lw> ow — Outer width oh — Outer height angle — Gap angle from 0 degrees  num_spokes — Number of spokes gap — Size of spoke gap lw — Line width

Symbol Name	Example	Parameters
Oblong Thermal		oblong_ths<ow>x<oh>x<angle>x<num_spokes>x<gap>x<lw>x<r s> ow — Outer width oh — Outer height angle — For 2 spokes, angle can be 0 or 90 degrees; for 4 spokes, angle can be 0, 45, or 90 degrees. num_spokes — Number of spokes gap — Size of spoke gap lw — Line width r s — Support for rounded or straight corners
Home Plate (hplate)		hplate<w>x<h>x<c> hplate<w>x<h>x<c>x<ra>x<ro> w — Horizontal side h — Vertical side c — Cut size (c<=w) ra — Corner radius (acute angle) (optional) ro — Corner radius (obtuse angle) (optional) ra and ro must meet the restrictions in “Symbols Suitable for Solder Stencil Design” on page 201.
Inverted Home Plate (rhplate)		rhplate<w>x<h>x<c> rhplate<w>x<h>x<c>x<ra>x<ro> w — Horizontal side h — Vertical side c — Cut size (c<w) ra — Corner radius (acute angle) (optional) ro — Corner radius (obtuse angle) (optional) ra and ro must meet the restrictions in “Symbols Suitable for Solder Stencil Design” on page 201.

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Symbol Name	Example	Parameters
Flat Home Plate (fhplate)	Symbol: fhplate2x1.5x0.7x0.5xra0.2xro0.1 	fhplate<w>x<h>x<vc>x<hc> fhplate<w>x<h>x<vc>x<hc>x<ra>x<ro> w — Horizontal side h — Vertical side hc — Horizontal cut size ($hc < w$) vc — Vertical cut size ($vc < h/2$) ra — Corner radius (acute angle) (optional) ro — Corner radius (obtuse angle) (optional) ra and ro must meet the restrictions in “Symbols Suitable for Solder Stencil Design” on page 201.
Radiused Inverted Home Plate (radhplate)	Symbol: radhplate0.5x0.5x0.15xra0.1 	radhplate<w>x<h>x<ms> radhplate<w>x<h>x<ms>x<ra> w — Horizontal side h — Vertical side ($h > (ms + (w - ms)/2)/2$) ms — Middle curve size ($ms < w$) ra — Corner radius (optional) ra and ro must meet the restrictions in “Symbols Suitable for Solder Stencil Design” on page 201.
Radiused Home Plate (dshape)	Symbol: dshape2x3x1xra0.003 	dshape<w>x<h>x<r> dshape<w>x<h>x<r>x<ra> w — Horizontal side h — Vertical side r — Relief ($r \leq w$ or ($r = w$ and $h = 2 * w$)) ra — Corner radius (optional) ra and ro must meet the restrictions in “Symbols Suitable for Solder Stencil Design” on page 201.
Cross (crossx[r[s]])	Symbol: cross3x3x1x1x50x50xra0.3 	The cross symbol consists of two intersecting orthogonal line segments. The cross point is at the intersection of their respective skeletons. cross<w>x<h>x<hs>x<vs>x<hc>x<vc>x[r[s]] cross<w>x<h>x<hs>x<vs>x<hc>x<vc>x[r[s]]<ra> w — Horizontal side h — Vertical side

Symbol Name	Example	Parameters
	Variations:  If the cross point is located outside the skeleton of the segment, the resulting indentation is ignored. The symbol is created like this  Not like this 	hs — Horizontal line width ($hs < h$) vs — Vertical line width ($vs < w$) hc — Horizontal cross point in percent vc — Vertical cross point in percent r s — Line style (round or square) ra — Inside corner radius (limited by the lengths of the line segments)
Dogbone (dogbonex[r s])	Symbol: <code>dogbone50x50x10x15x50xsxra3</code>  Variation: 	dogbone<w>x<h>x<hs>x<vs>x<hc>x[r s] dogbone<w>x<h>x<hs>x<vs>x<hc>x[r s]x<ra> w — Horizontal side h — Vertical side hs — Horizontal line width ($hs < h/2$) vs — Vertical line width ($vs < w$) hc — Horizontal cross point r s — Line style (round/square) ra — Corner radius (must be less than the whichever value is smaller: hs or $vs/2$)
D-Pack (dpack)	Symbol: <code>dpack100x100x15x15x3x3</code> 	The individual pads of dpack can be straight or rounded, depending on the corner radius (ra) parameter. dpack<w>x<h>x<hg>x<vg>x<hn>x<vn> dpack<w>x<h>x<hg>x<vg>x<hn>x<vn>x<ra> w — Horizontal side h — Vertical side hg — Horizontal gap ($hg < h$) vg — Vertical gap ($vg < v$) vn — Columns number

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Symbol Name	Example	Parameters
	Symbol: <code>dpack100x100x15x15x3x3xral</code>	hn — Rows number ra — Corner radius ($ra \leq [\text{the smaller of } hs \text{ and } vs]/2$) The horizontal and vertical sides of an individual pad within the D-Pack are calculated. They are not user-defined. They must be greater than zero. $hs = w - hg \cdot (hn - 1) / hn$ $vs = h - (vg \cdot (vn - 1)) / vn$
Ellipse	Symbol : <code>el60x30</code>	el<w>x<h> w — Width h — Height
Moire	Symbol : <code>moire5x10x4x4x100x0</code>	moire<rw>x<rg>x<nr>x<lw>x<ll>x<la> rw — Ring width rg — Ring gap nr — Number of rings lw — Line width ll — Line length la — Line angle
Hole	Symbol : <code>hole50xp4x5</code>	hole<d>x<p>x<tp>x<tm> d — Hole diameter p — Plating status (p(lated), n(on-plated) or v(ia)) tp — Positive tolerance tm — Negative tolerance Intended for wheels created for drill files.